

CS/ECE 752 Spring 2025: HOMEWORK # 1
(Due by **12:30 PM** on **Friday, Feb 7**, via upload of PDF to Canvas)

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Problem 1 (4 points)

Assume that a processor (multicycle design) takes 1 cycle for integer instructions (except multiply), 3 cycles for loads/stores, 4 cycles for branches, and 6 cycles for multiply. Assume that a program executes 40% integer instructions, 15% multiplies, 30% loads/stores, and 15% branches.

- (a) Determine the CPI of the program.
- (b) If 75% of the multiply instructions can be converted into 2 shifts and 1 add instruction, what is the new CPI of the program?

Problem 2 (4 points)

Figure 1 gives a comparison of power and performance for several benchmarks comparing two servers: Sun Fire T2000 (which uses Niagara) and IBM x346 (using Intel Xeon processors).

- (a) Calculate the performance/power ratio for each processor on each benchmark.
- (b) If power is your main concern, which would you choose? Why?

	Sun Fire T2000	IBM x346
Power (watts)	298	438
SPECjbb (op/s)	63,378	39,985
Power (watts)	330	438
SPECWeb (composite)	14,001	4,348

Figure 1. Sun power / performance comparison as selectively reported by Sun

Problem 3 (4 points)

- (a) Assume that 40% of a processor can be sped up by 12X. What is the overall speedup of the processor?
- (b) What is the overall speedup if 40% of the processor is sped up by 200X.

Comment on your observations.

Problem 4 (12 points)

Make the following calculations on the raw data in order to explore how different measures color the conclusions one can make. (Doing these exercises will be much easier using a spreadsheet.)

- (a) Create a table similar to that shown in Figure 2, except express the results as normalized to the Pentium 4 for each benchmark.
- (b) Calculate the arithmetic mean of the performance (Memory and Dhrystone performance) of each processor. Use both the original performance and your normalized performance calculated in part (a).
- (c) Given your answer from part (b), can you draw any conflicting conclusions about the relative performance of the different processors?
- (d) Calculate the geometric mean of the normalized performance of the dual processors and the geometric mean of the normalized performance of the single processors for the Dhrystone benchmark.

Chip	# of cores	Clock frequency (MHz)	Memory performance	Dhrystone performance
Athlon 64 X2 4800+	2	2,400	3,423	20,718
Pentium EE 840	2	2,200	3,228	18,893
Pentium D 820	2	3,000	3,000	15,220
Athlon 64 X2 3800+	2	3,200	2,941	17,129
Pentium 4	1	2,800	2,731	7,621
Athlon 64 3000+	1	1,800	2,953	7,628
Pentium 4 570	1	2,800	3,501	11,210
Processor X	1	3,000	7,000	5,000

Figure 2. Performance of several processors on two benchmarks.

Problem 5 (4 points)

Your company has a CPU that takes 4 cycles to execute a MUL instruction, 6 cycles to execute a DIV instruction, and 1 cycle to execute all other instructions. There is no pipelining. The target workload has an instruction mix containing 25% MUL instructions, 15% DIV instructions, and 60% others.

- (a) What is the Cycles Per Instruction (CPI) of this CPU running this target workload?
- (b) An improvement in the divider makes the DIV instruction take 1 cycle to execute, but causes the overall cycle time to increase by 30%. Does this new design make the workload run faster than the original CPU? Explain.

Problem 6 (12 points)

For this question you will need to do additional exploration and research; we did not cover this in class.

Processors with different instruction sets (ISAs) are commonly used in different computing environments. Consider the following different application domains. For each domain: (i) what is the dominant ISA?, (ii) which companies are the predominant manufacturers of the processors in the domain?, (iii) what is the approximate volume (order of magnitude) of processors sold worldwide in the domain? Discuss the implications of your analysis.

(i) Data Centers such as those run by Amazon, Meta, Google, Microsoft

(ii) Smartphones and Tablets

(iii) Laptops and PCs

(iv) RISC-V (<https://riscv.org/>) is a recent open source instruction set (ISA) that is targeting many different application domains. The rationale is that RISC-V is an open source ISA, and thus anybody can use it to design chips for their application space, without being constrained by, or having to pay expensive licensing fees for proprietary intellectual property. What opportunities and challenges do you see for RISC-V in the Datacenters domain?